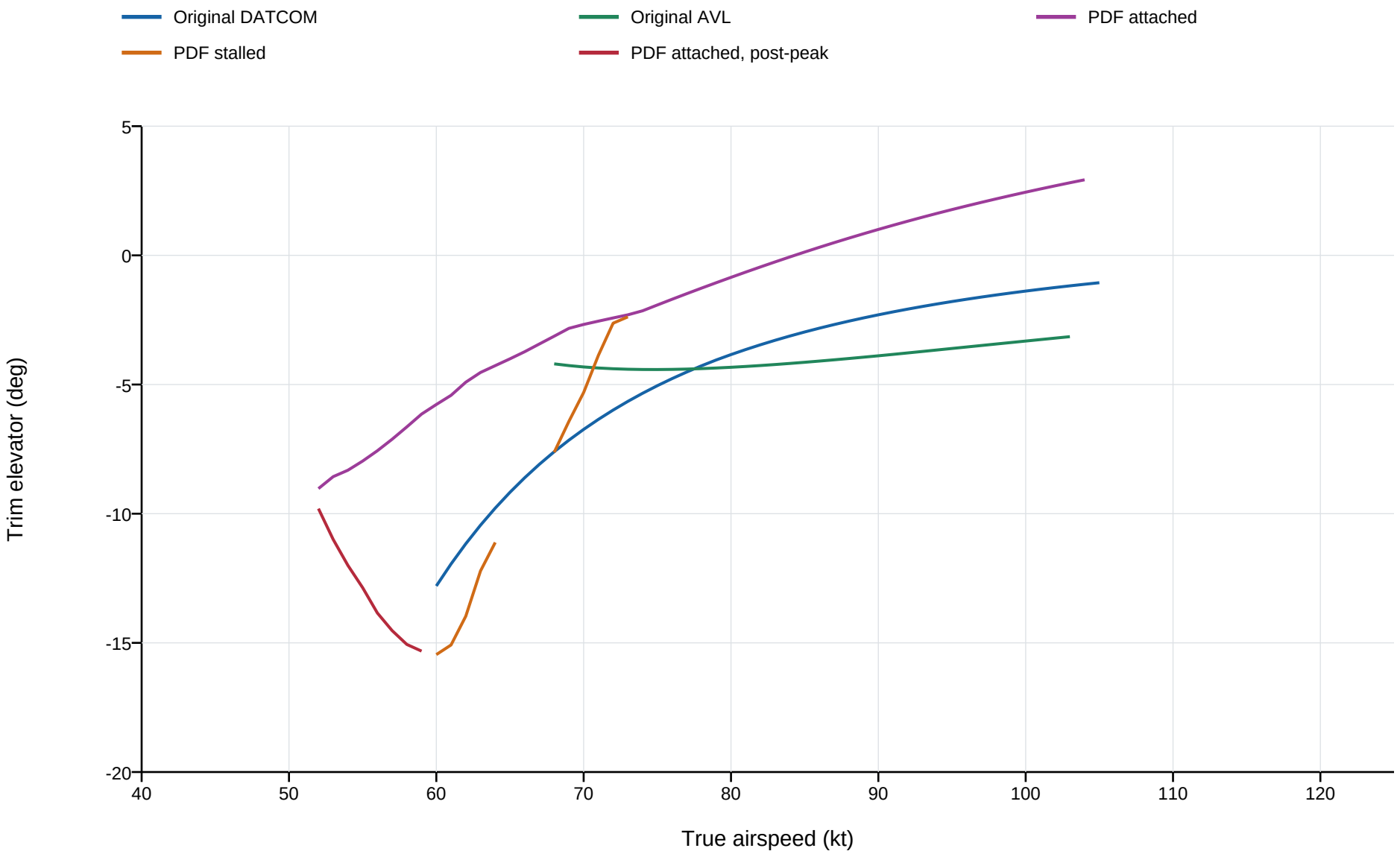
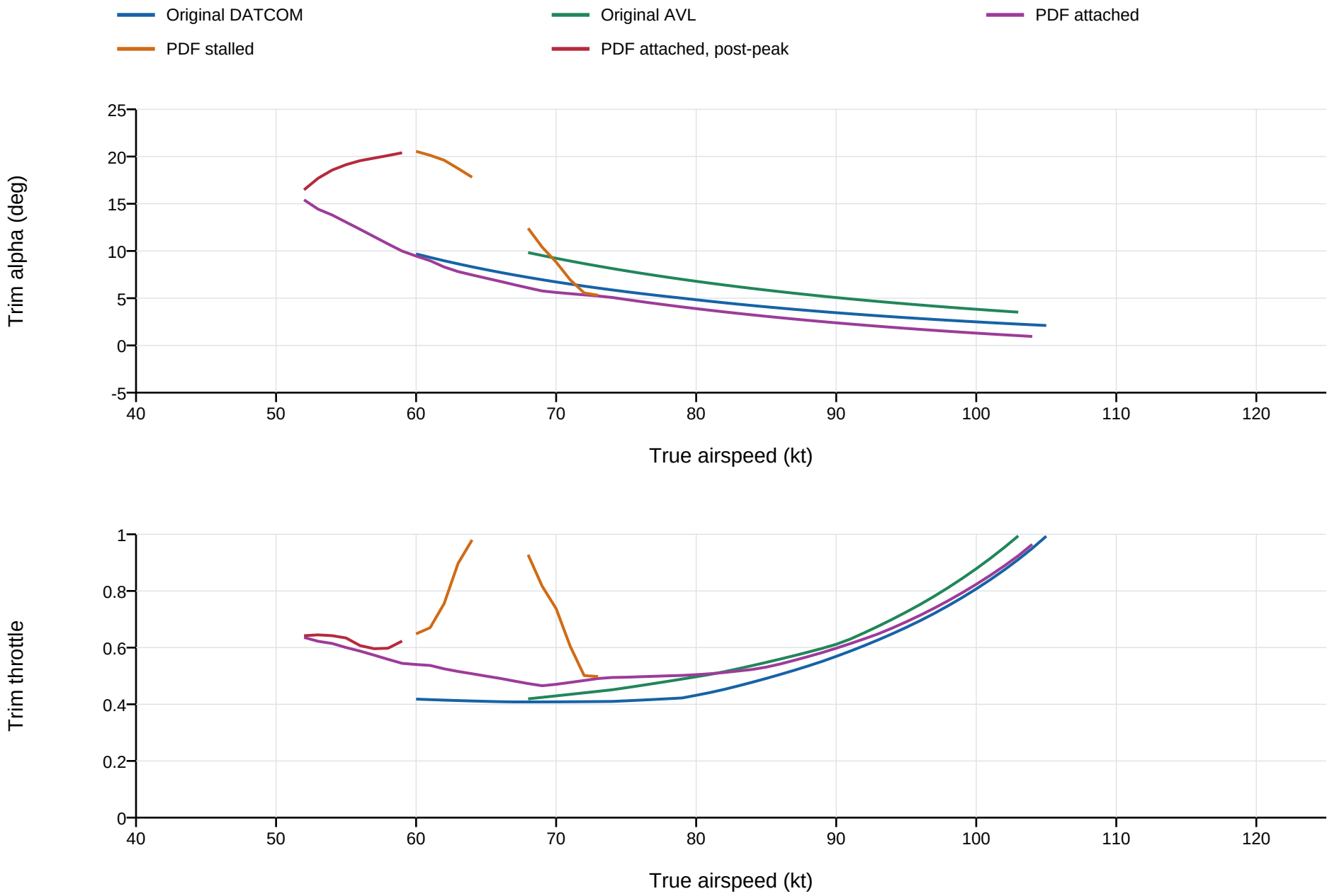
Trim elevator versus airspeed

Straight level flight; sea-level density; clean wing; mass 1043 kg; common propeller model.
Airspeed 40-125 kt in 1 kt steps; zero body rates; only roots with throttle between 0 and 1 are plotted.



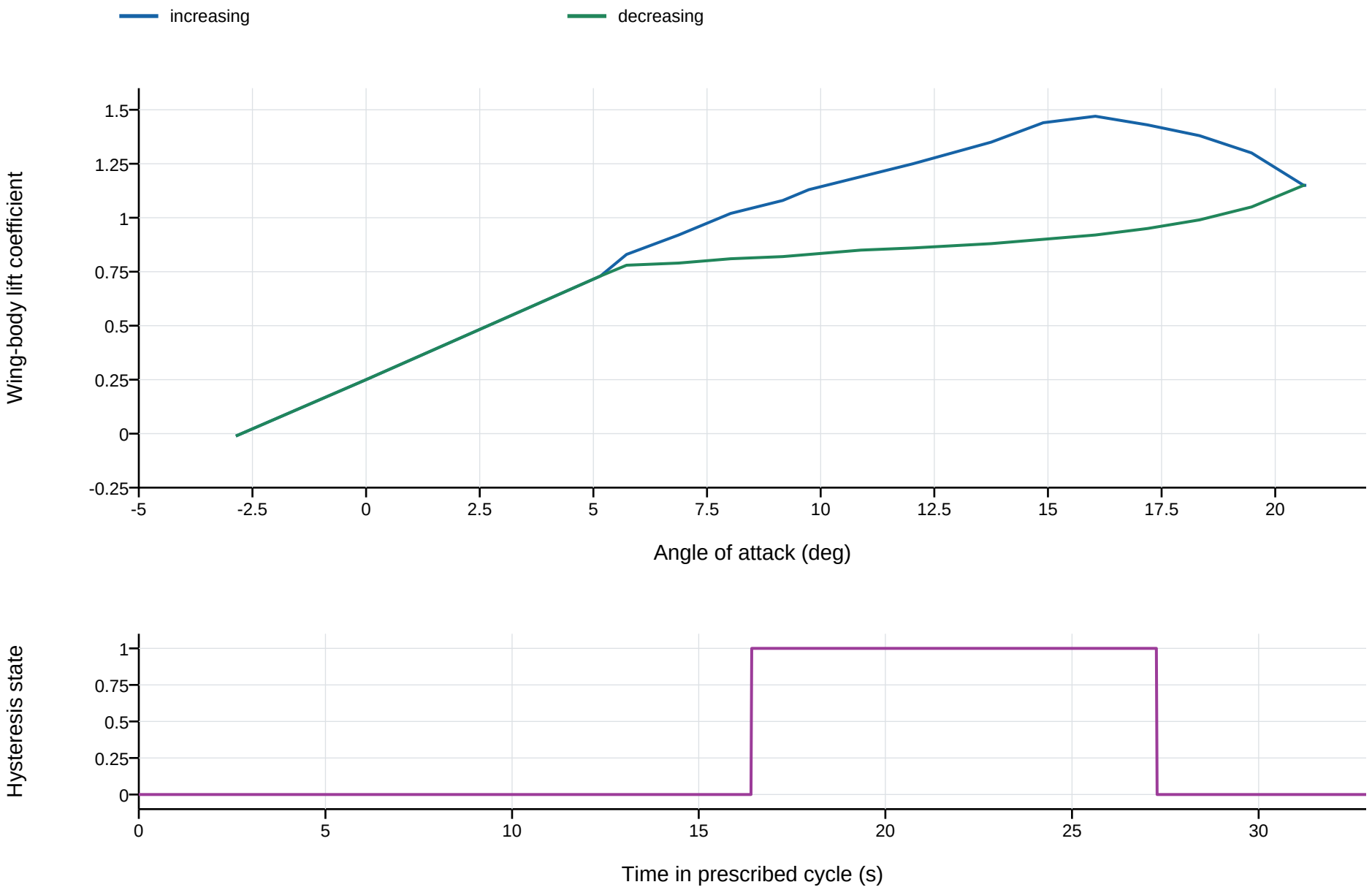
Trim angle of attack and throttle

Same level-flight cases and constraints as the elevator comparison.



Increasing and decreasing alpha: stall memory

Prescribed alpha cycle: -0.05 to 0.361 rad and back; sampling interval 0.02 s.
Hysteresis sets above 0.36 rad and resets below 0.09 rad; wing-body CL is plotted.



6DOF time response to elevator input

Each model starts at its own 100 kt level-flight trim; sea-level density; duration 15 s; timestep 0.01 s.
Additional elevator -1 deg from 2 to 3 s; trim throttle held fixed; clean configuration.

